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Monte Carlo and Resampling Methods Applied to Weak Gravitational Lensing Mass Reconstruction

MA 551 - Computational Statistics

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SECTION 01

Motivation

Why lensing? Why statistics?

01

Weak Gravitational Lensing

Massive structures bend light from background galaxies, distorting their apparent shapes coherently across the sky.

The observable: galaxy ellipticity $e = e_1 + ie_2$

What we want: projected mass density $\kappa(\boldsymbol{\theta})$ (convergence)

- Direct probe of dark matter, dark energy equation of state
- Stage-IV surveys (Rubin, Euclid, Roman) are already underway
- Systematic errors at the **sub-percent level** will limit cosmological inference

The core problem: how do we propagate uncertainty, detect bias, and determine how much data we need?

The Statistical Problem

The mass reconstruction pipeline has two stages:

1. **Forward model:** $\kappa \to \gamma$ via a known linear FFT operator F
2. **Inverse:** recover $\hat{\kappa}$ from noisy shear observations γ_{obs}

Each source galaxy contributes an ellipticity measurement with intrinsic scatter $\sigma_e = 0.26$ per component (“shape noise”):

$$\gamma_{\text{obs}[i,j]} = (1+m)\gamma_{\text{true}[i,j]} + \varepsilon[i,j], \quad \varepsilon[i,j] \sim \mathcal{N}\left(0, \sigma_e^2 / n_{\text{gal/pix}}\right)$$

Key questions (all from MA 551):

- What is the reconstruction error, and how variable is it? **(MC)**
- Can we quantify uncertainty efficiently? **(Antithetic, IS)**
- What confidence intervals are valid? **(Bootstrap, BCa)**
- How many galaxies to detect 1% bias at 5σ? **(Power analysis)**

SECTION 02

The Kaiser–Squires Algorithm

A validated R implementation

02

Kaiser–Squires Inversion

Shear and convergence are related in Fourier space via the KS kernel $D(\boldsymbol{k})$. Since $|D(\boldsymbol{k})| = 1$ for all $\boldsymbol{k} \neq \mathbf{0}$, the inverse is simply the conjugate. Optional Tikhonov regularization: replace $|D|^2$ with $|D|^2 + \lambda$.

$$\hat{\kappa}(\boldsymbol{k}) = \overline{D(\boldsymbol{k})}\hat{\gamma}(\boldsymbol{k}), \quad D(\boldsymbol{k}) = \frac{k_1^2 - k_2^2 + 2ik_1k_2}{k_1^2 + k_2^2}$$

DC ambiguity: $D(\mathbf{0}) = 0$ always, so the mean of κ is unrecoverable (mass-sheet degeneracy). The noiseless KS aperture mass (0.250) differs from the true value (0.346) by 9.6% due to this irreducible offset.

Analytical Validation: 10-Test Suite

The R implementation was validated against SMPy's `KaiserSquiresMapper` (production Python KS). Both compute the same algebraic expression.

Test	Result
FFT frequency convention matches `numpy.fft.fftfreq`	PASS
$\lvert D(\boldsymbol{k})\rvert = 1$ for all $\boldsymbol{k} \neq \mathbf{0}$ (machine precision)	PASS
Noiseless round-trip L_2 error $< 10^{-10}$	PASS
Linearity: $F(a\kappa_1 + b\kappa_2) = aF(\kappa_1) + bF(\kappa_2)$	PASS
Parseval / adjoint: $\langle F\kappa, \gamma\rangle = \langle \kappa, F^*\gamma\rangle$	PASS
B-mode purity $< 10^{-10}$ for physical κ	PASS
SMPy exact formula equivalence: max diff $< 10^{-14}$	PASS
Single-frequency (sinusoidal) analytical solution	PASS
DC non-contamination: adding constant to κ does not change γ	PASS
Peak amplitude recovered to 10^{-10} (DC-corrected)	PASS

SECTION 03

Simulation Study

Monte Carlo, antithetic variables, importance sampling

03

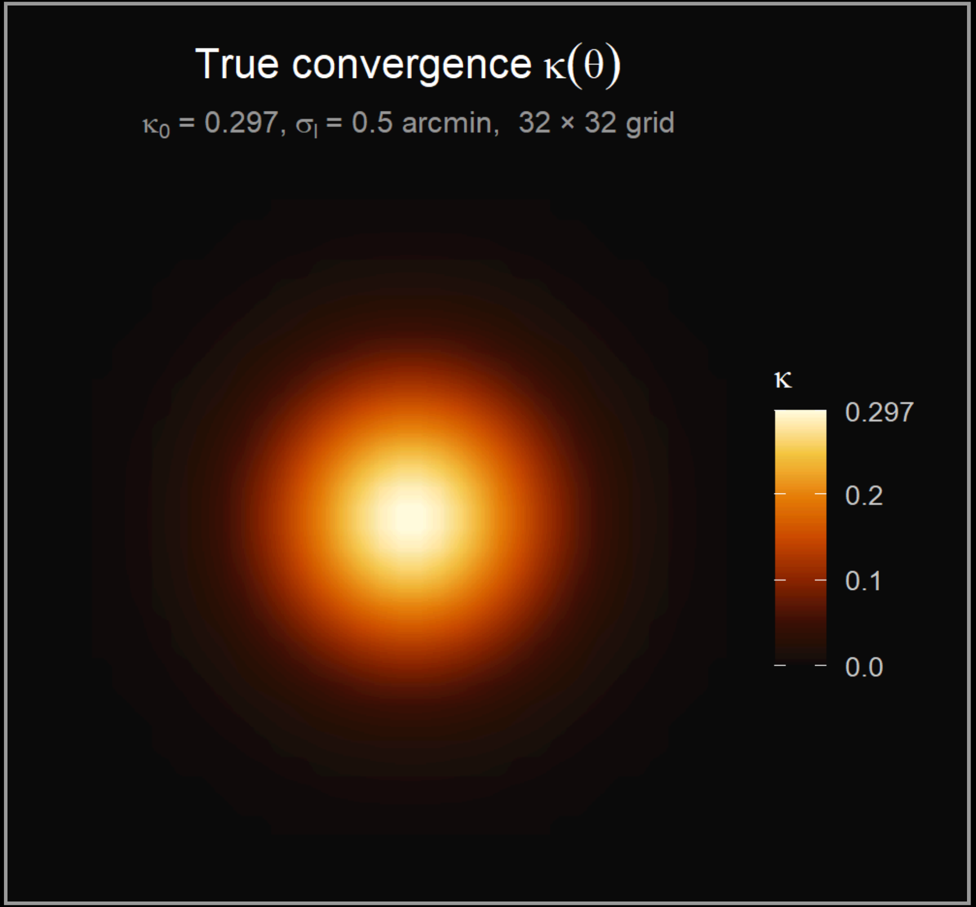
Simulation Setup

Grid: 32×32 pixels, 0.1 arcmin/pixel
Field: 3.2×3.2 arcmin^2
True κ: Gaussian cluster, peak $= 0.297$, $\sigma_\ell = 0.5$ arcmin
Galaxies: $N_{\rm gal} = 5,000$
Noise: $\sigma_{\rm pix} = 0.118$, max $|\gamma| = 0.103$
SNR per pixel: ≈ 0.85 (noise-dominated)

Two scalar summaries:

- Peak $\hat{\kappa}$: maximum pixel value
- Aperture mass: $T_{\rm ap} = \sum_{r \leq 0.8} \hat{\kappa}\ \delta\theta^2$

Aperture mass is preferred: it integrates over a region rather than taking the maximum of a noisy field.



Simple Monte Carlo: $B = 500$ Replicates

Estimate $\mathbb{E}[L_2(m=0)]$ and the distribution of scalar summaries under pure shape noise (Notes #12).

Summary	True value	MC mean	MC SD
L_2 error	0 $(m = 0)$	1.521	0.045
Peak $\hat{\kappa}$	0.297	**0.488**	0.051
Aperture mass	0.250†	0.251	0.015

† Noiseless KS value; true-κ aperture mass is 0.346 (DC offset 9.6%).

- **Peak kappa is severely upward-biased:** the maximum of 1024 noisy pixels is a noise spike, not the cluster. Bootstrap is not valid for this statistic.
- **Aperture mass is nearly unbiased** relative to the correct target (noiseless KS).
- L_2 error mean of 1.52 reflects the noise-dominated regime at $N_{\text{gal}} = 5000$.

Antithetic Variable Sampling (Notes #13)

Pair each noise draw ε with $-\varepsilon$. For monotone g:

$$\mathrm{Var}\left(\hat{I}_{\mathrm{anti}}\right) = \frac{\sigma^2}{2N}(1 + \mathrm{Cor}(g(\varepsilon), g(-\varepsilon))) < \frac{\sigma^2}{2N}$$

Statistic	**Variance reduction**	**Reason**
Aperture mass	**+100%**	Linear in ε; $\mathrm{Cor} = -1$ exactly
Peak $\hat{\kappa}$	-4%	Maximum is not monotone
L_2 error	-100%	**Degenerate** (see below)

The degenerate case: $L_2(\varepsilon) = \|F^{-1}\varepsilon\| \,/\, \|\kappa_{\mathrm{true}}\|$ depends only on $\|\varepsilon\|$, so $L_2(\varepsilon) = L_2(-\varepsilon)$ **exactly.** Antithetic pairs are identical; averaging them halves the effective sample size without any variance reduction.

This illustrates the key condition from Notes #13: **antithetic sampling requires the integrand to be monotone.** An even function in ε violates this.

Importance Sampling Over Bias (Notes #12, #13)

Estimate the bias-integrated error:

$$\mathbb{E}_{p[L_2(m)]} = \int L_2(m)p(m), dm, \quad p(m) = \mathcal{N}(0, 0.05^2)$$

Proposal: $q(m) = \mathrm{Uniform}(-0.15, 0.15)$. Self-normalized weights $w_i = \frac{p(m_i)}{q(m_i)}$.

Method	**Estimate (SE)**
IS ($n = 50$, $B' = 100$)	1.518 (0.193)
Uniform grid	1.520

The IS and uniform estimates agree closely. This is itself informative: **the reconstruction error is nearly flat in m at this galaxy density.** The bias signal is buried in shape noise, so IS offers no efficiency gain. A more informative regime (higher N_{gal}) would show larger IS benefit near $m = 0$ where the prior concentrates.

SECTION 04

Bootstrap Inference

BCa confidence intervals and coverage

04

Parametric Bootstrap Setup (Notes #6, #7)

Parametric bootstrap: draw $B = 500$ fresh noise realizations from the known $\mathcal{N}(0, \sigma^2_{\text{pix}})$ distribution. The noise model is known exactly.

BCa correction factors:

- **Bias correction** $\hat{z}_0 = \Phi^{-1}\left(B^{-1} \sum_b \mathbb{1}\left(\hat{T}^*_b < \hat{T}\right)\right)$: adjusts for median shift
- **Acceleration** $\hat{a}$: estimated via **column jackknife** on the shear grid. Delete column j, recompute KS reconstruction. Requires $N = 32$ reconstructions (not N^2). For near-linear estimators, $\hat{a} \approx 0$.

Statistic	Obs.	Bias	SE	95% BCa CI
Peak $\hat{\kappa}$	0.251	+0.237	0.051	degenerate
Aperture mass	0.250	+0.000	0.015	$[0.220, 0.278]$

Peak BCa fails: every bootstrap replicate adds fresh noise, so the bootstrap peak **always exceeds** the noiseless reference. $\hat{z}_0 = \Phi^{-1}(0) = -\infty$.

Coverage Study: $n_{\text{outer}} = 200$ Replicates

For each outer replicate: generate one noisy observation, compute BCa and percentile CI, check coverage against the noiseless KS aperture mass target.

Interval type	Empirical coverage	Mean width
Percentile	97.5%	0.0586
BCa	97.5%	0.0585

Interpretation:

- Both achieve 97.5% at nominal 95%: slightly conservative, as expected when σ_{pix} is known exactly and the noise is exactly Gaussian.
- BCa $\approx$ percentile here because $\hat{a} = 0.006 \approx 0$: the aperture mass functional is nearly linear in the shear, so the estimator's SE is essentially constant and no skewness correction is needed.
- BCa would diverge from percentile for non-linear statistics (e.g., peak kappa) or at lower SNR where asymmetry becomes important.

SECTION 05

Power Analysis

How many galaxies to detect 1% bias at 5σ?

05

Design: Why Unpaired?

Goal: detect multiplicative bias $|m| = 0.01$ via two-sample Welch t-test on B independent aperture mass replicates per condition.

The paired design is degenerate for linear statistics.

Since aperture mass is linear in the shear, the paired difference is:

$$D_b = T_{\mathrm{ap}(m=0.01)} - T_{\mathrm{ap}(m=0)} = 0.01 \times T_{\mathrm{ap}}^{\mathrm{KS}} = \mathrm{const}$$

Every replicate gives exactly the same D_b, so $\mathrm{sd}(D) = 0$ and NCP $= \infty$. This is the same linearity that caused the antithetic degeneracy.

Design: Why Unpaired?

Unpaired design: independent noise in each condition. Effect size is analytic:

$$\delta = |m| \times T_{\mathrm{ap}}^{\mathrm{KS}} = 0.01 \times 0.2504 = 0.00250$$

SD scales as $\frac{1}{\sqrt{N_{\mathrm{gal}}}}$ from the shape noise model:

$$\sigma_{\mathrm{ap}(N_{\mathrm{gal}})} = 0.0146 \times \sqrt{\frac{5000}{N_{\mathrm{gal}}}}$$

Power Curve: Detecting $|m| = 0.01$ at 5σ

Non-centrality parameter and power for $B = 500$ replicates per condition, significance threshold $\alpha = 2.87 \times 10^{-7}$ (5-sigma).

N_{gal}	NCP	Power	σ_{ap}
1,000	1.21	< 0.001	0.033
2,000	1.71	0.001	0.023
5,000	2.70	0.008	0.015
10,000	3.83	0.096	0.010
20,000	5.41	0.610	0.007
50,000	**8.55**	$\mathbf{> 0.999}$	**0.005**

Minimum N_{gal} for 80% power: $\approx$ 24,400 (analytic uniroot).

MC verification at $N_{\text{gal}} = 5000$, $B = 500$: NCP $= 2.30$, power $= 0.002$ (analytic: 2.70, 0.008 – 15% gap from finite-B fluctuation, as expected).

SECTION 06

Conclusions

06

Summary of Findings

Method	**Key finding**
Simple MC	Aperture mass nearly unbiased; peak kappa severely biased by noise spikes
Antithetic	+100% reduction for aperture mass; **degenerate** for L_2 error (even function)
IS	No gain here: error flat in m at $N_{\text{gal}} = 5000$; bias buried in noise
Bootstrap	Aperture mass BCa well-behaved ($\hat{a} \approx 0$); peak kappa BCa undefined
Coverage	97.5% empirical at nominal 95%; BCa $\approx$ percentile for linear functionals
Power	24,400 galaxies for 80% power at 5σ detecting $\|m\| = 0.01$, $B = 500$

Unifying theme: the linearity of the KS operator creates a consistent pattern. Linear statistics (aperture mass) are tractable for bootstrap and IS but degenerate for antithetic and paired designs. Non-linear statistics (peak) are biased and bootstrap-invalid. Statistical method choice is inseparable from the structure of the estimator.

Connections to MA 551 Curriculum

Every analysis in this project maps directly to course material:

- **Notes #3–4:** simulation-based hypothesis testing, critical values
- **Notes #6:** the bootstrap, empirical CDF, resampling
- **Notes #7:** BCa CI, jackknife, acceleration factor
- **Notes #12:** Monte Carlo integration, simple MC estimation
- **Notes #13:** antithetic variables (monotonicity condition), importance sampling (proposal design), control variates

The project is approximately 80% statistical analysis and 20% forward modeling. The astrophysics provides a concrete inverse problem where the statistical methods produce physically interpretable results.

Future work: GalSim-simulated galaxy images with realistic PSF convolution and metacalibration shear response correction, placing this analysis in a physically complete pipeline.

Acknowledgements

George Vassilakis Open-source Kaiser--Squires reference implementation.

Thank you for your time!

Questions?